Gaussian Belief Propagation for the Diffusion Score

The fully solvable AR(1)+OU chain: joint score, factor graph, sum–product, Kalman, closed-form bulk analysis, and the breakdown of AMP

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June 10, 2026

Abstract

We study the *joint* score of a diffusion model on a sequence of correlated frames in the only setting where everything can be carried to the end in closed form: a stationary Gaussian AR(1) Markov chain whose frames are independently corrupted by a variance-preserving Ornstein–Uhlenbeck channel. The note is written to be self-contained: we derive the forward channel from the OU stochastic differential equation, prove the three Gaussian lemmas used by every later computation, derive the clean joint law and its tridiagonal precision, the noised law $P_t(x) = \mathcal{N}(0, \Sigma_t)$, and the joint score $S(x,t) = -\Sigma_t^{-1}x$ both directly and through Tweedie's identity, whose proof we give in full together with its denoiser, expected-noise, and tilted-measure readings and the reverse-time SDE in which the score is the drift. The $K = 2$ case is worked out completely as the smallest example of inter-frame coupling. We then write the posterior $p(a \,|\, x)$ as a factor graph (a caterpillar tree), prove it is a tree, derive belief propagation from scratch with an explicit bookkeeping convention that avoids the double-counting trap, execute every message by hand on the $K = 3$ chain — symbolically and numerically — and identify the sweeps with the Kalman filter and RTS smoother. Gaussianity of all messages is an *exact algebraic closure*, not a central limit effect. New to this version, we solve the chain in closed form in its homogeneous bulk: the exact posterior covariance is geometric, $(J^{-1})_{i,i+d} = q^{\,d}/\sqrt{J_\mathrm{d}^2 - 4\beta^2}$, the BP message precision has the explicit fixed point $\lambda^* = (J_\mathrm{d} + \sqrt{J_\mathrm{d}^2 - 4\beta^2})/2$ which exists for *every* coupling, and the AMP/TAP variance closure has the explicit fixed point $V = (J_\mathrm{d} - \sqrt{J_\mathrm{d}^2 - 8\beta^2})/4\beta^2$ which exists *iff* $J_\mathrm{d} \geq 2\sqrt{2}\,|\beta|$ — giving an exact breakdown time $t_c(\alpha)$ and a critical coupling $\alpha_c = \sqrt{2} - 1$ below which AMP never breaks down. BP, AMP, and mean field share the same posterior mean, hence the *same score*; they differ only in the variance, where AMP first deviates at fourth order in the coupling, $V_{\text{AMP}} - V_{\text{exact}} = 2\beta^4/J_\mathrm{d}^5 + \ldots$, and then ceases to exist — making quantitative J. Garnier-Brun's remark that a central limit theorem "on two points" need not hold. Two experiments close the note: the locality of the score (the truncation error of a radius-r estimator decays exactly as $q^{\,r}$), and the lifecycle of the precision matrix Σ_t^{-1} along diffusion time, including the corrected band-fill law $(Q_t)_{i,i+d} = (-1)^{d-1}(2t)^{d-1}(Q_0^d)_{i,i+d} + O(t^d)$ (no $1/(d-1)!$ factor). Every quantitative claim is verified by an independent numerical audit (**72/72 checks pass**); three executed companion notebooks reproduce every number and figure.

Contents

1 Introduction and motivation

What generative diffusion is. Generative diffusion aims at producing artificial data $a \in \mathbb{R}^d$ distributed according to a law $P_0(a)$ that is not known explicitly but from which we hold n examples. The recipe has two halves: *noising* the data by a diffusion process until all structure is destroyed, then *denoising* by a reverse-time diffusion whose drift contains a trained vector field — the *score* $\nabla_x \log P_t(x)$ [7, 11, 12]. This is the state of the art in image and video generation, and the same trained score also gives likelihood estimates of new data points.

Our setting: dynamic objects. This project studies diffusion on *dynamic objects*: ordered sequences with an internal time, the canonical example being a short video clip. A dynamic object is a sequence of K frames $a_0, \dots, a_{K-1}$, drawn from a clean joint distribution with *Markov* structure,

$$P_0(a_0, \dots, a_{K-1}) = p_0(a_0) \prod_{u=0}^{K-2} M(a_{u+1} \,|\, a_u), \tag{1}$$

and the diffusion paradigm corrupts each frame independently with an Ornstein–Uhlenbeck (OU) process at diffusion time t. Sampling from P_0 then requires the *joint score field*

$$S(x,t) := \nabla_x \log P_t(x_0, \dots, x_{K-1}) \in \mathbb{R}^K. \tag{2}$$

Remark 1 (Joint, not marginal — the central correction)**.** The object of interest is the score of the *joint* law of the whole sequence. It is emphatically *not* the per-frame marginal score $\nabla_{x_k} \log p_t(x_k)$, which integrates out every other frame and would be the right object only if the frames were generated independently. A video is *one* sample of a joint law, not K samples of a marginal; the joint score couples each frame to the entire trajectory through the full posterior $P(a \,|\, x)$. Every formula below is a joint-score formula. (For the model of this note the per-frame marginal score is the trivial $-x_k$ at every t — see §6 — so the marginal formulation would erase precisely the structure we want to study.)

Why the Gaussian ar(1) chain. We take the simplest nontrivial instance of (1): a stationary Gaussian AR(1) chain of scalar frames. Three reasons make it the right testbed.

1. **Everything is closed form.** The clean law, the noised law, the posterior, the score, the message-passing fixed points, even the breakdown boundary of AMP — all are explicit formulas, each verified independently.
2. **It separates the two structural mechanisms.** Gaussianity is responsible for the *linearity* of the score; Markovianity is responsible for the *tridiagonal sparsity* of the clean precision. The literature often conflates them; here each can be traced separately.
3. **It is the cleanest stage for the algorithmic question.** The posterior is a tree-structured graphical model, so belief propagation (BP) is exact and $O(K)$; the same model viewed as a Gaussian Markov random field lets us run the AMP/TAP closure side by side and see precisely where, and why, the dense-graph approximation fails on a chain.

Questions this note answers. The note grew out of working sessions with M. Mézard and J. Garnier-Brun and is organised around their questions:

- Derive the clean joint, the noised joint, and the score — both directly and "the Bayesian way" (Tweedie). Do the two agree? (§5: yes, identically; audited.)
- Write the posterior, draw its factor graph, run BP. Are the messages Gaussian? Is the posterior Gaussian? Is BP exact here? (§10: yes, yes, yes — by algebraic closure and tree-exactness, *not* by any central limit argument.)
- Apply AMP. Is AMP the same as BP here? (§12: the *score* is the same — the mean solves the same linear system — but the variance closures differ, and AMP's has no fixed point beyond an explicit boundary.)
- What is the difference between running strictly local messages and the full forward–backward sweeps? (§13: the truncation error decays exactly as q^r with an explicit $q(\alpha, t)$.)
- How does the tridiagonal structure of the precision die and come back along diffusion time? (§14: the band-fill law, the spectral flow, and the return to the identity, all explicit.)

New closed-form results. Beyond consolidating the derivations, this version contains five results we have not seen stated in this form elsewhere (all proved in §11–§12 and verified by the audit):

1. the exact *bulk posterior covariance* of the chain, $(J^{-1})_{i,i+d} = q^d/\sqrt{J_\mathrm{d}^2 - 4\beta^2}$, with $q = (J_\mathrm{d} - \sqrt{J_\mathrm{d}^2 - 4\beta^2})/2|\beta|$;
2. the BP message-precision fixed point $\lambda^* = (J_\mathrm{d} + \sqrt{J_\mathrm{d}^2 - 4\beta^2})/2$, which exists for *every* (α, t) — BP never breaks down on the chain;
3. the AMP/TAP variance fixed point $V = (J_\mathrm{d} - \sqrt{J_\mathrm{d}^2 - 8\beta^2})/4\beta^2$ and its existence condition $J_\mathrm{d} \geq 2\sqrt{2}|\beta|$, hence an exact breakdown time $t_c(\alpha)$ and the critical coupling $\alpha_c = \sqrt{2} - 1 \approx 0.4142$;
4. the weak-coupling law $V_\mathrm{AMP} - V_\mathrm{exact} = 2\beta^4/J_\mathrm{d}^5 + O(\beta^6)$: AMP's variance is correct to second order in the coupling and first errs — always upward — at fourth;
5. the locality law: the RMS truncation error of the radius-r local estimator decays exactly as q^r, with the same q as in (i).

Methodological rule. Every numbered claim in this note is backed by a named check in `code/numerical_audit.py`. A statement enters the text only if its check passes; the audit currently passes **72/72**. We prefer fewer results that are certainly right over more that are merely plausible. Appendix A maps claims to checks.

How to read this note. Sections 2–3 fix the model and the Gaussian toolbox. Sections 4–6 derive the laws and the score, ending with the fully explicit $K = 2$ case. Sections 7–10 build the factor graph and belief propagation, with $K = 3$ done entirely by hand. Sections 11–12 contain the closed-form bulk analysis and the BP/AMP comparison. Sections 13–14 are the two experiments. Section 15 collects what is established and what comes next. Three executed notebooks (`01_model_and_score`, `02_belief_propagation`, `03_amp_and_experiments`) mirror the structure of the note.

2 Setup, notation, and the OU forward channel

2.1 The model

A *sequence* is K scalar frames $a = (a_0, \dots, a_{K-1}) \in \mathbb{R}^K$, indexed by the internal (frame) time $k = 0, \dots, K-1$. The frames are generated by a stationary Gaussian AR(1) Markov chain,

$$a_{k+1} = \alpha\, a_k + \eta_k, \qquad \eta_k \sim \mathcal{N}(0, \sigma_\eta^2) \text{ i.i.d.}, \qquad |\alpha| < 1, \qquad a_0 \sim \mathcal{N}(0, \sigma_0^2). \tag{3}$$

Throughout we use the *stationary, unit-variance* normalisation

$$\sigma_\eta^2 = 1 - \alpha^2, \qquad \sigma_0^2 = \frac{\sigma_\eta^2}{1 - \alpha^2} = 1, \tag{4}$$

so that $\mathrm{Var}(a_k) = 1$ for every k (proved in §4). The scalar setting $d = 1$ loses nothing: every construction below acts per component.

2.2 The OU channel, derived from its SDE

Each frame is corrupted *independently* by the variance-preserving OU process

$$\mathrm{d}x = -x\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W, \qquad x(0) = a_k. \tag{5}$$

Proposition 1 (The forward channel)**.** *At diffusion time $t \geq 0$ the solution of* (5) *started at a_k is the affine Gaussian channel*

$$x_k = e^{-t} a_k + \xi_k, \qquad \xi_k \sim \mathcal{N}(0, \Delta_t), \qquad \Delta_t := 1 - e^{-2t}, \tag{6}$$

independent across k.

Derivation 1: solving the linear SDE

Multiply (5) by the integrating factor e^t: $\mathrm{d}(e^t x) = e^t(\mathrm{d}x + x\,\mathrm{d}t) = \sqrt{2}\,e^t \mathrm{d}W$, so

$$x(t) = e^{-t} a_k + \sqrt{2} \int_0^t e^{-(t-s)}\,\mathrm{d}W_s.$$

The stochastic integral is Gaussian with mean zero and variance (Itô isometry)

$$2 \int_0^t e^{-2(t-s)}\,\mathrm{d}s = 1 - e^{-2t} = \Delta_t.$$

Hence $x_k \sim \mathcal{N}(e^{-t} a_k, \Delta_t)$, and the channels are independent across k because each frame is driven by its own Brownian motion.

Reading : variance preservation and the two clocks

At $t = 0$, $x_k = a_k$ (no corruption); as $t \to \infty$, $x_k \to \mathcal{N}(0, 1)$ (pure noise). In between, $\mathrm{Var}(x_k) = e^{-2t} \cdot 1 + \Delta_t = 1$ *for all* t: the channel trades signal for noise at constant total variance. Two clocks run through the whole note and must never be confused: the *frame* index k (internal time of the sequence) and the *diffusion* time t (noise level).

2.3 Notation table

k	frame index $(0, \dots, K-1)$	t	diffusion (noise) time
a_k	clean frame	x_k	noisy observation of a_k at time t
α	$\textsc{ar}(1)$ coupling	$\sigma_\eta^2 = 1-\alpha^2$	innovation variance
$\mu := e^{-t}$	channel attenuation	$\Delta_t = 1 - e^{-2t}$	channel noise variance
$\Sigma_0,\ \Sigma_t$	clean / noised covariance	$Q_0,\ Q_t$	their inverses (precisions)
$S(x,t)$	**joint** score $\nabla_x \log P_t(x)$	$\langle\cdot\rangle_{a\,\|\,x}$	posterior expectation
$J,\ h$	posterior precision / field (§7)	$m = J^{-1}h$	posterior mean
$J_{\mathrm{d}},\ \beta$	bulk diagonal / off-diagonal of J (§11)	q	posterior correlation decay (§11)
λ, θ	information form of a Gaussian message	ω_i	eigenvalues of Σ_0

3 The Gaussian toolbox

Three elementary lemmas carry every computation in this note. We state and prove them once, in the parametrisation we will actually use.

Information form. A one-dimensional Gaussian in a will often be written

$$\mu(a) \ \propto\ \exp\bigl(-\tfrac{1}{2}\lambda a^2 + \theta a\bigr) \qquad \Longleftrightarrow \qquad \text{variance } v = \tfrac{1}{\lambda}, \quad \text{mean } m = \tfrac{\theta}{\lambda}. \tag{7}$$

λ is the *precision*, θ the *potential.* The point of this form: multiplication of densities becomes *addition* of (λ, θ) pairs.

Lemma 2 (Product of Gaussians)**.** $\mathcal{N}(a; m_1, v_1) \cdot \mathcal{N}(a; m_2, v_2) \ \propto\ \mathcal{N}(a; m, v)$ *with*

$$\frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \qquad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2} \qquad \textit{(precisions and potentials add).}$$

Derivation 2

In information form the two factors are $\exp(-\frac{1}{2}\lambda_i a^2 + \theta_i a)$ with $\lambda_i = 1/v_i$, $\theta_i = m_i/v_i$. Their product is $\exp\bigl(-\frac{1}{2}(\lambda_1+\lambda_2)a^2 + (\theta_1+\theta_2)a\bigr)$, again of the form (7), with the stated parameters. The product of two Gaussian *densities* is an unnormalised Gaussian: the normaliser is irrelevant for message passing, where every message is defined up to a constant.

Lemma 3 (Gaussian propagation / predict step)**.**

$$\int \mathcal{N}(b; \alpha a, s^2)\, \mathcal{N}(a; m, v)\, \mathrm{d}a \ = \ \mathcal{N}\bigl(b;\ \alpha m,\ \alpha^2 v + s^2\bigr).$$

Derivation 3

The integral computes the law of $b = \alpha a + \eta$ with $a \sim \mathcal{N}(m, v)$ and $\eta \sim \mathcal{N}(0, s^2)$ independent. A linear combination of independent Gaussians is Gaussian; its mean is αm and its variance $\alpha^2 v + s^2$ by independence. (No completing-the-square gymnastics is needed once the integral is recognised as a marginalisation of a joint Gaussian.)

Lemma 4 (Quadratic form $\Rightarrow$ Gaussian; vector case)**.** *If* $p(a) \propto \exp\bigl(-\frac{1}{2}a^\top J a + h^\top a\bigr)$ *on* $\mathbb{R}^K$ *with J symmetric positive definite, then* $p = \mathcal{N}(a;\ J^{-1}h,\ J^{-1})$.

Derivation 4: completing the square

With $m := J^{-1}h$,

$$-\tfrac{1}{2}a^\top Ja + h^\top a = -\tfrac{1}{2}(a-m)^\top J(a-m) + \tfrac{1}{2}m^\top Jm,$$

as expanding the right side confirms. The constant term joins the normalisation, leaving the kernel of $\mathcal{N}(m, J^{-1})$.

Reading : why these three lemmas are the whole story

Belief propagation on our model only ever (i) multiplies two Gaussians in the same variable (Lemma 2: combination steps), (ii) pushes a Gaussian through the AR(1) transition (Lemma 3: predict steps), and (iii) reads off a mean and covariance from a quadratic log-density (Lemma 4: the posterior). Each operation maps Gaussians to Gaussians *exactly*. This algebraic closure — not any large-N or central-limit argument — is what will make every BP message Gaussian in §10.

4 The clean joint distribution

4.1 Autocovariance and stationarity

Unrolling (3) from a_0, $a_k = \alpha^k a_0 + \sum_{j=0}^{k-1} \alpha^{k-1-j}\eta_j$.

Proposition 5 (Stationary covariance)**.** *With the normalisation* (4), $\mathbb{E}[a_k] = 0$ *and*

$$(\Sigma_0)_{ij} := \operatorname{Cov}(a_i, a_j) = \alpha^{|i-j|} \qquad \textit{for all } i, j. \tag{8}$$

Derivation 5: variance, then covariance

Variance. Since $a_0 \perp \eta_j$ and the η_j are i.i.d.,

$$\operatorname{Var}(a_k) = \alpha^{2k}\sigma_0^2 + \sigma_\eta^2 \sum_{j=0}^{k-1} \alpha^{2(k-1-j)} = \alpha^{2k}\sigma_0^2 + \sigma_\eta^2 \frac{1-\alpha^{2k}}{1-\alpha^2}.$$

With $\sigma_0^2 = \sigma_\eta^2/(1-\alpha^2)$ the two terms combine to $\sigma_\eta^2/(1-\alpha^2) = 1$ for *all* k: the chain is stationary from frame 0. (For $|\alpha| = 1$ the variance grows linearly — random walk; for $|\alpha| > 1$ exponentially; we keep $|\alpha| < 1$.)
Covariance. For $i \geq j$, unroll $a_i = \alpha^{i-j}a_j + \sum_l \alpha^{\cdots}\eta_l$ where every η_l in the sum is indexed at or after j and is therefore independent of a_j. Hence $\mathbb{E}[a_i a_j] = \alpha^{i-j}\mathbb{E}[a_j^2] = \alpha^{i-j}$. By symmetry, $\operatorname{Cov}(a_i, a_j) = \alpha^{|i-j|}$.

4.2 Markov factorisation and the tridiagonal precision

The joint density factorises by the chain rule and the Markov property,

$$P_0(a) = p_0(a_0) \prod_{k=0}^{K-2} M(a_{k+1} \,|\, a_k) = \mathcal{N}(a_0; 0, 1) \prod_{k=0}^{K-2} \mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2). \tag{9}$$

Proposition 6 (Tridiagonal clean precision)**.** $Q_0 := \Sigma_0^{-1}$ *is exactly tridiagonal:*

$$(Q_0)_{kk} = \begin{cases} \dfrac{1}{\sigma_\eta^2}, & k \in \{0, K-1\}, \\ \dfrac{1+\alpha^2}{\sigma_\eta^2}, & 0 < k < K-1, \end{cases} \qquad (Q_0)_{k,k\pm 1} = -\frac{\alpha}{\sigma_\eta^2},$$

and all entries at distance ≥ 2 *vanish identically.*

Derivation 6: from the log-density

From (9),

$$-2\log P_0(a) = \frac{a_0^2}{\sigma_0^2} + \sum_{k=0}^{K-2} \frac{(a_{k+1} - \alpha a_k)^2}{\sigma_\eta^2} + \text{const},$$

and Q_0 is the Hessian of $-\log P_0$ (Lemma 4). Off-diagonals: only the adjacent transition term couples a_k to a_{k+1}, contributing the cross-coefficient $-\alpha/\sigma_\eta^2$; no term couples variables at distance ≥ 2. Interior diagonal: a_k appears in the left transition $(a_k - \alpha a_{k-1})^2$ with coefficient $1/\sigma_\eta^2$ and in the right transition $(a_{k+1} - \alpha a_k)^2$ with coefficient α^2/σ_η^2, summing to $(1+\alpha^2)/\sigma_\eta^2$. Left boundary: the prior contributes $1/\sigma_0^2 = (1-\alpha^2)/\sigma_\eta^2$ and the right transition α^2/σ_η^2; the sum is $1/\sigma_\eta^2$. The right boundary is symmetric (only a left transition).

Reading : tridiagonality is Markovianity, not Gaussianity

Σ_0 is *dense* (geometrically decaying Toeplitz); its inverse Q_0 is *tridiagonal.* A zero in the precision is a conditional independence given all the rest, and the chain couples only neighbours. The derivation used Gaussianity only to identify the precision with the Hessian of $-\log P_0$; the *support* of that Hessian — first neighbours only — is purely the chain factorisation (9). Any Markov chain with smooth transitions has this sparsity pattern in $\nabla^2(-\log P_0)$; the Gaussian case just makes the entries constants. This tridiagonal Q_0 is the only sparse object in the entire story, and every structural statement below traces back to it.

5 The noisy joint distribution and the score, two ways

The channel (6) acts independently on each frame, so given the clean sequence,

$$p_t(x \mid a) = \prod_{k=0}^{K-1} \mathcal{N}(x_k; e^{-t} a_k, \Delta_t), \tag{10}$$

and the noised joint law marginalises over the clean sequence:

$$P_t(x) = \int_{\mathbb{R}^K} P_0(a)\, p_t(x \mid a)\, \mathrm{d}a. \tag{11}$$

Proposition 7 (Noised covariance)**.** $P_t(x) = \mathcal{N}(x; 0, \Sigma_t)$ *with*

$$\boxed{\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I_K} \tag{12}$$

Derivation 7

With $x = e^{-t}a + \xi$, $a \sim \mathcal{N}(0, \Sigma_0)$, $\xi \sim \mathcal{N}(0, \Delta_t I)$ independent: $\mathbb{E}[x] = 0$ and $\mathrm{Cov}(x) = e^{-2t}\,\mathrm{Cov}(a) + \mathrm{Cov}(\xi) = e^{-2t}\Sigma_0 + \Delta_t I$. A linear map plus an independent Gaussian applied to a Gaussian is Gaussian, so the law is fully determined by these two moments.

5.1 The score, directly

Since $\log P_t(x) = -\frac{1}{2}x^\top \Sigma_t^{-1} x + \text{const}$,

$$\boxed{S(x,t) = \nabla_x \log P_t(x) = -\Sigma_t^{-1}x = -Q_t\, x} \qquad Q_t := \Sigma_t^{-1}. \tag{13}$$

The score is *linear* in x (Gaussianity), and its coupling structure is exactly the precision Q_t: every structural question about the joint score is a question about a matrix. Sections 11 and 14 describe this matrix completely.

5.2 The score, the Bayesian way: Tweedie's identity

Theorem 8 (Joint Tweedie identity [3, 7, 12])**.** *For any prior P_0 on $\mathbb{R}^K$ (not only Gaussian) and the channel (10), the joint score admits, component-wise, the two equivalent forms*

$$\boxed{S_k(x,t) = \frac{e^{-t}\,\mathbb{E}[a_k \,|\, x] - x_k}{\Delta_t} = -\frac{\mathbb{E}[\xi_k \,|\, x]}{\sqrt{\Delta_t}}} \tag{14}$$

where the expectations are under the posterior $P(a \,|\, x)$ and ξ is the standardised channel noise, $x = e^{-t}a + \sqrt{\Delta_t}\,\xi$.

Derivation 8: from the channel to Tweedie, in full

Step 1 (log-gradient). For any positive f, $\nabla \log f = \nabla f / f$; applied to P_t, the problem is to compute $\partial_{x_k} P_t(x)$ and divide by $P_t(x)$.
Step 2 (differentiate under the integral). The prior does not depend on x, so from (11)

$$\partial_{x_k} P_t(x) = \int P_0(a)\, \partial_{x_k} p_t(x \,|\, a)\, \mathrm{d}a.$$

The likelihood is an explicit Gaussian in x_k: $\log p_t(x \,|\, a) = -\frac{(x_k - e^{-t}a_k)^2}{2\Delta_t} + (x_k\text{-indep.})$, so

$$\partial_{x_k} p_t(x \,|\, a) = -\frac{x_k - e^{-t}a_k}{\Delta_t}\; p_t(x \,|\, a).$$

Step 3 (Bayes). Substituting and splitting the integral,

$$\partial_{x_k} P_t(x) = -\frac{1}{\Delta_t}\Big[x_k\, P_t(x) - e^{-t} \int a_k\, P_0(a)\, p_t(x \,|\, a)\, \mathrm{d}a\Big].$$

The joint density of (a, x) is $P_0(a)p_t(x \,|\, a)$; dividing by its marginal $P_t(x)$ gives the posterior, $P(a \,|\, x) = P_0(a)p_t(x \,|\, a)/P_t(x)$. Hence dividing the display by $P_t(x)$,

$$S_k(x,t) = \frac{\partial_{x_k} P_t(x)}{P_t(x)} = -\frac{x_k - e^{-t}\,\mathbb{E}[a_k \,|\, x]}{\Delta_t},$$

which is the first form in (14).

Step 4 (noise form). On the joint space $\xi_k = (x_k - e^{-t}a_k)/\sqrt{\Delta_t}$. Conditioning on x (under which x_k is a constant) and taking expectations, $\mathbb{E}[\xi_k \,|\, x] = (x_k - e^{-t}\mathbb{E}[a_k \,|\, x])/\sqrt{\Delta_t} = -\sqrt{\Delta_t}\, S_k(x,t)$, the second form. $\square$

Bayesian view : three readings of one identity

(i) The score is the optimal denoiser. Solving (14),

$$\mathbb{E}[a \,|\, x] = \frac{x + \Delta_t\, S(x,t)}{e^{-t}} :$$

knowing the score and knowing the Bayes-optimal denoiser is the *same* problem, related by an affine, prior-independent rescaling. **(ii) The score is the posterior expected noise** (second form): what the reverse-time SDE adds back during sampling is, literally, the expected corruption. **(iii) The prior enters only through** $\mathbb{E}[a \,|\, x]$: every structural property of the score — linearity, locality, factorisation — is inherited from the Bayesian posterior mean. This third reading organises the rest of the note: computing the joint score *is* computing a posterior mean, and the algorithms of §10 are posterior-mean algorithms.

Bayesian view : the tilted measure — how much to trust the data

For a single component (the argument is component-wise), completing the square in a inside Bayes' rule gives

$$\mathbb{E}[a \,|\, x] = \frac{1}{Z_x}\int a\, p_0(a)\, \exp\Big(-\frac{e^{-2t}}{2\Delta_t}\,(a-\widetilde{x})^2\Big)\mathrm{d}a, \qquad \widetilde{x} := e^t x.$$

The posterior mean is the prior average *tilted* by a quadratic pinning centred at the deconvolved observation $\widetilde{x}$, with pinning strength e^{-2t}/Δ_t [7]. Small t: the strength diverges like $1/2t$, the tilt concentrates at $\widetilde{x} \to x$, the score vanishes — the data is trusted. Large t: the strength vanishes, the tilt flattens, $\mathbb{E}[a \,|\, x] \to 0$ (prior mean) and $S \to -x$ — the data is ignored. All the nontrivial structure of diffusion lives at intermediate t, where pinning and prior compete. We will meet this crossover again, quantitatively, in the bulk parameters of §11.

Derivation : where the score enters generation — the reverse-time SDE

For completeness we record why the score is the central object [7, 11]. The forward OU process has Fokker–Planck equation $\partial_t P_t = \nabla\cdot(xP_t) + \Delta P_t$. Set $\tau := t_f - t$ and $\widetilde{P}_\tau := P_{t_f-\tau}$; then $\partial_\tau \widetilde{P}_\tau = -\nabla\cdot(x\widetilde{P}_\tau) - \Delta\widetilde{P}_\tau$, which is *not* the FPE of any diffusion (the Laplacian has the wrong sign). Using $\Delta P = \nabla\cdot\big((\nabla \log P)P\big)$ and the identity $-A = -2A + A$ on the Laplacian term,

$$\partial_\tau \widetilde{P}_\tau = -\nabla\cdot\Big[\big(x + 2\nabla\log\widetilde{P}_\tau\big)\widetilde{P}_\tau\Big] + \Delta\widetilde{P}_\tau,$$

which *is* the FPE of the reverse-time SDE

$$\mathrm{d}x = \big(x + 2\,S(x,\, t_f - \tau)\big)\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}\widetilde{W}_\tau.$$

Generation = run this SDE from pure noise; the only unknown is the score. In our model the drift is the explicit linear field $x + 2S = \big(I - 2Q_t\big)x$ at each noise level.

Reading : the two forms agree — and why that is a real check

Equations (13) and (14) compute the same vector through completely different routes: a $K \times K$ matrix inversion versus a Bayesian posterior mean. For the Gaussian model the posterior mean is linear in x (§7), and substituting it into (14) must return $-Q_t x$ identically. The audit verifies the two routes agree to 2.7×10^{-15} across (K, α, t) sweeps — a nontrivial end-to-end consistency check of every formula derived so far.

6 $K = 2$: the smallest example of inter-frame coupling

Before any algorithm, it is worth seeing every object of §4–5 written out where nothing can hide. With $K = 2$ and unit stationary variance,

$$\Sigma_0 = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}, \qquad \Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \boxed{r := \alpha\, e^{-2t}}$$

(the diagonal is $e^{-2t} + \Delta_t = 1$: variance preservation; the off-diagonal is the clean correlation α attenuated by e^{-2t}, because the *noise* of the two frames is independent and only the *signal* correlates). Inverting the 2×2 matrix,

$$Q_t = \frac{1}{1 - r^2}\begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}, \qquad S_0(x,t) = -\frac{x_0 - r\,x_1}{1 - r^2}, \qquad S_1(x,t) = -\frac{x_1 - r\,x_0}{1 - r^2}. \tag{15}$$

Reading : what the joint score knows that the marginal does not

The per-frame *marginal* score is trivial here: each x_k is $\mathcal{N}(0,1)$ at every t (variance preservation), so $\nabla_{x_k} \log p_t(x_k) = -x_k$, *independent of t and of the other frame.* The joint score (15) instead contains the cross term $-r\,x_1$ in S_0: the denoiser of frame 0 *borrows evidence* from frame 1, with a weight $r = \alpha e^{-2t}$ that is the entire coupling lifecycle in miniature — equal to α at $t = 0$, decaying exponentially, gone at $t \to \infty$. This is the structure that the marginal formulation erases (Remark 1), and the reason the joint score is the object of study.

Numerical check : $K = 2$ closed forms

`numerical_audit.py` checks (15) against the generic machinery (Σ_t assembly and numerical inversion) over 16 configurations $\alpha \in \{0.3, 0.7, 0.95, -0.5\}$, $t \in \{0.01, 0.2, 1, 4\}$ and 80 random scores: maximum errors 1.1×10^{-16} (Σ_t), 2.2×10^{-16} (Q_t), 1.8×10^{-15} (score).

7 The posterior and its factor graph

7.1 The posterior in closed form

By Bayes, the posterior over the clean sequence given the (fixed) observations is

$$p(a \,|\, x) \;\propto\; P_0(a)\, p_t(x \,|\, a) = \underbrace{\mathcal{N}(a_0; 0, 1)}_{\psi_{\mathrm{pr}}(a_0)} \prod_{k=0}^{K-2} \underbrace{\mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2)}_{\psi_{\mathrm{tr}}^{(k)}(a_k, a_{k+1})} \prod_{k=0}^{K-1} \underbrace{\mathcal{N}(x_k; e^{-t} a_k, \Delta_t)}_{\psi_{\mathrm{ob}}^{(k)}(a_k)}. \tag{16}$$

Proposition 9 (Gaussian posterior)**.**

$$p(a \mid x) = \mathcal{N}(a;\, m,\, J^{-1}), \qquad \boxed{J = \frac{e^{-2t}}{\Delta_t} I + Q_0, \qquad m = J^{-1}h, \qquad h = \frac{e^{-t}}{\Delta_t} x} \tag{17}$$

Derivation 9: collecting the quadratic form

Take $-\log$ of (16) and collect powers of a. The prior and transitions contribute $\frac{1}{2}a^\top Q_0 a$ (Proposition 6). Each observation factor contributes

$$\frac{(x_k - e^{-t}a_k)^2}{2\Delta_t} = \frac{e^{-2t}}{2\Delta_t} a_k^2 - \frac{e^{-t}}{\Delta_t} x_k\, a_k + (a\text{-indep.}),$$

i.e. $\frac{e^{-2t}}{\Delta_t}$ to the diagonal of the quadratic form and $\frac{e^{-t}}{\Delta_t} x_k$ to the linear term. Lemma 4 then gives (17). Note that J is *tridiagonal* (Q_0 tridiagonal plus a multiple of the identity) and that the posterior mean $m = J^{-1}h$ is *linear* in x, as promised below Theorem 8. Substituting m into Tweedie (14) and simplifying recovers $S = -Q_t x$, which is the end-to-end consistency the audit checks.

7.2 The marginalisation problem: where the cost lives

To evaluate S_k via (14) we need the marginal posterior mean $\mathbb{E}[a_k \mid x]$, i.e. the one-dimensional marginal of (16). Direct integration,

$$p(a_k \mid x) \;\propto\; \int \cdots \int \Big[\psi_{\mathrm{pr}} \textstyle\prod_u \psi_{\mathrm{tr}}^{(u)} \prod_j \psi_{\mathrm{ob}}^{(j)} \Big]\, \mathrm{d}a_{-k},$$

costs $K-1$ one-dimensional integrals (each variable is eliminated by an integral that involves only its neighbours' factors), and repeating this for every k costs $K(K-1) = O(K^2)$ — or $O(K^3)$ if one instead inverts J blindly. The across-marginal waste is evident: the integral over a_j is recomputed for every target $k \neq j$. Belief propagation stores each intermediate integral as a reusable *message* and brings the total to $2(K-1)$ integrals: this — not any approximation — is the entire point of BP on this model.

7.3 Factor graph

A *factor graph* is the bipartite graph with one node per variable, one node per factor, and an edge whenever the factor depends on the variable [6, 8]. For (16): the variables are $a_0, \ldots, a_{K-1}$; the factors are 1 prior + $(K-1)$ transitions + K observations = $2K$ in total. The observations x_k are *not* variable nodes — they are constants clamped inside $\psi_{\mathrm{ob}}^{(k)}$.

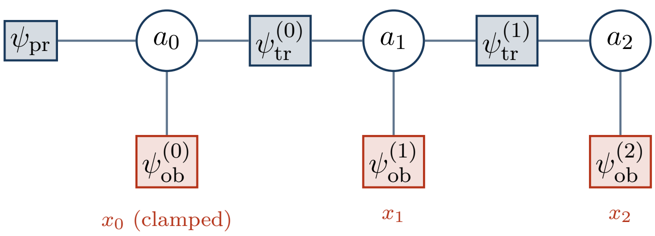

Figure 1: Factor graph of $p(a \mid x)$ for $K = 3$. Circles: variable (clean frame) nodes. Blue squares: prior and transition factors. Red squares: observation factors carrying the clamped data x_k. The graph is a *caterpillar tree*: a backbone a_0–a_1–a_2 with one leaf factor per node.

Proposition 10 (The graph is a tree)**.** *The factor graph of* (16) *is connected and acyclic.*

Derivation 10: counting edges

Nodes: K variables $+2K$ factors $= 3K$. Edges: each transition factor has degree 2 ($2K-2$ edges), each observation factor degree 1 (K edges), the prior degree 1 (1 edge): total $3K-1$ = nodes -1. The graph is connected (every a_k is reachable from a_0 along the transition backbone), and a connected graph with $V-1$ edges is a tree.

Proposition 11 (Global Markov property)**.** *Conditioning on* a_k *makes the left block* $(a_0,\ldots,a_{k-1};x_0,\ldots,x_{k-1})$ *and the right block* $(a_{k+1},\ldots,a_{K-1};x_{k+1},\ldots,x_{K-1})$ *independent: every path between the blocks passes through* a_k*, and conditioning severs it.*

Reading : this is exactly the geometry J. Garnier-Brun described

The graph is "very tree-like — a straight line with things on the side." Every interior variable has degree three (two transitions, one observation); every BP update will combine a *small, fixed* number of incoming messages. Two consequences, one per direction of the note: (i) BP is *exact* on trees [8, Th. 14.1] — no message ever returns to its sender, so nothing is double-counted; (ii) this is the *opposite* of the dense, high-degree graphs for which AMP was built (§12), which is where the BP/AMP contrast will come from.

8 Belief propagation on the chain

8.1 The sum–product rules

On a generic factor graph, sum–product BP passes two kinds of message along every edge — variable-to-factor and factor-to-variable [6, 8, 9]:

$$\nu_{i\to a}(a_i) \;\propto\; \prod_{b\in\partial i\setminus a} \hat\nu_{b\to i}(a_i), \tag{18}$$

$$\hat\nu_{a\to i}(a_i) \;\propto\; \int \psi_a(a_{\partial a}) \prod_{j\in\partial a\setminus i} \nu_{j\to a}(a_j)\,\mathrm{d}a_{\partial a\setminus i}, \tag{19}$$

$$p(a_i) \;\propto\; \prod_{a\in\partial i} \hat\nu_{a\to i}(a_i). \tag{20}$$

On a tree these equations have a unique solution reached in one inward and one outward pass, and the beliefs (20) are the *exact* marginals.

8.2 Convention A on the chain, and the double-counting trap

On the chain the rules collapse into one *forward* and one *backward* sweep. The bookkeeping deserves care — an earlier internal version of these notes fell into a trap here, which the audit caught.

Proposition 12 (Convention A)**.** *Define messages (functions of the receiving variable)*

$$\mu_{\to 0}(a_0) := \psi_{\mathrm{pr}}(a_0), \tag{21}$$

$$\mu_{\to k+1}(a_{k+1}) := \int \psi_{\mathrm{tr}}^{(k)}(a_k, a_{k+1})\, \psi_{\mathrm{ob}}^{(k)}(a_k)\, \mu_{\to k}(a_k)\, \mathrm{d}a_k, \tag{22}$$

$$\mu_{\leftarrow K-1}(a_{K-1}) := 1, \tag{23}$$

$$\mu_{\leftarrow k-1}(a_{k-1}) := \int \psi_{\mathrm{tr}}^{(k-1)}(a_{k-1}, a_k)\, \psi_{\mathrm{ob}}^{(k)}(a_k)\, \mu_{\leftarrow k}(a_k)\, \mathrm{d}a_k. \tag{24}$$

Then, for every k,

$$p(a_k \mid x) \;\propto\; \mu_{\to k}(a_k)\; \psi_{\mathrm{ob}}^{(k)}(a_k)\; \mu_{\leftarrow k}(a_k). \tag{25}$$

The forward message into a_k carries the evidence $x_0, \ldots, x_{k-1}$ only; the backward message carries $x_{k+1}, \ldots, x_{K-1}$ only; the local evidence $\psi_{\mathrm{ob}}^{(k)}$ enters once, at combination time.

Derivation 12: peeling the chain at a_k

Integrate the joint (16) over every a_j, $j \neq k$. By the global Markov property (Proposition 11) the integral splits at a_k into independent left and right halves; the local factor $\psi_{\mathrm{ob}}^{(k)}(a_k)$ depends on no integrated variable and factors out. **Left half:** define $\mu_{\to k}(a_k) := \int \psi_{\mathrm{pr}} \prod_{u<k} \psi_{\mathrm{tr}}^{(u)} \prod_{j<k} \psi_{\mathrm{ob}}^{(j)}\, \mathrm{d}a_0 \cdots \mathrm{d}a_{k-1}$. Peeling the innermost variable a_{k-1} — only three factors involve it — yields exactly the recursion (22), with base case (21). **Right half:** the mirror argument gives (24)–(23); the base case is flat because there is no evidence to the right of the last frame. **Combination:** left integral $\times$ local evidence $\times$ right integral is (25), with normaliser $P_t(x)$.

Remark 2 (The double-counting trap)**.** A natural-looking alternative (*Convention B*) absorbs the local evidence into the forward message, $\mu^{\mathrm{B}}_{\to k} := \mu^{\mathrm{A}}_{\to k} \cdot \psi_{\mathrm{ob}}^{(k)}$. Under Convention B the simple product (25) counts x_k *twice* — once inside the message, once at combination — and must be corrected by dividing one factor back out. Convention A is the bookkeeping under which the textbook combination rule is correct *as written*. The audit was decisive in pinning this down: under Convention A the BP score matches the matrix score to 10^{-14}; under naive Convention B the disagreement is order one. This is the single most error-prone point of the whole construction, and the reason every formula in this section is stated with its evidence bookkeeping made explicit.

8.3 Complexity

Each sweep computes $K-1$ messages, each by a single one-dimensional integral; the combination at each node is a product of three one-variable functions. Total: $2(K-1)$ integrals + K combinations $= O(K)$, against $O(K^2)$ for direct per-marginal elimination and $O(K^3)$ for matrix inversion. For the Gaussian model each "integral" is two arithmetic operations on (λ, θ) pairs, so the constant is tiny as well.

9 BP on the $K = 3$ chain: every message, by hand

We now execute (21)–(25) on Fig. 1, first symbolically, then numerically on a concrete instance. Every message is Gaussian (the integrals are instances of Lemmas 2 and 3), so each is carried by a (mean, variance) pair.

9.1 The local evidence as a Gaussian in a_k

Read as a function of a_k, the observation factor is

$$\psi_{\mathrm{ob}}^{(k)}(a_k) = \mathcal{N}(x_k; e^{-t}a_k, \Delta_t) \;\propto\; \exp\Big(-\tfrac{1}{2}\tfrac{e^{-2t}}{\Delta_t}\,a_k^2 + \tfrac{e^{-t}}{\Delta_t}x_k\,a_k\Big) \;\Longleftrightarrow\; \mathcal{N}\big(a_k;\ e^t x_k,\ \Delta_t e^{2t}\big), \tag{26}$$

a Gaussian *pseudo-observation* of a_k at the deconvolved location $e^t x_k$ with variance $\Delta_t e^{2t}$ (precision e^{-2t}/Δ_t: the pinning strength of the tilted-measure box in §5).

9.2 Forward sweep = predict–update filter

Derivation : forward sweep (Convention A)

Initialisation. $\mu_{\to 0} = \psi_{\mathrm{pr}} = \mathcal{N}(a_0; 0, 1)$.
Generic step $k \to k+1$. Two half-steps:

- *Update* (Lemma 2): combine the incoming message with the local evidence (26),

$$\mathcal{N}(m_k^{\to}, v_k^{\to})\times\mathcal{N}(e^t x_k, \Delta_t e^{2t}) \;\longrightarrow\; \mathcal{N}(m_k^c, v_k^c), \qquad \frac{1}{v_k^c} = \frac{1}{v_k^{\to}} + \frac{e^{-2t}}{\Delta_t}, \qquad \frac{m_k^c}{v_k^c} = \frac{m_k^{\to}}{v_k^{\to}} + \frac{e^{-t}}{\Delta_t}x_k.$$

- *Predict* (Lemma 3): push through the transition,

$$\mu_{\to k+1} = \int \mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2)\,\mathcal{N}(a_k; m_k^c, v_k^c)\,\mathrm{d}a_k = \mathcal{N}\big(a_{k+1};\ \alpha m_k^c,\ \alpha^2 v_k^c + \sigma_\eta^2\big).$$

This is precisely the predict–update cycle of the *Kalman filter* [5]: BP on a Gaussian chain *is* Kalman filtering.

Derivation : backward sweep

Start flat, $\mu_{\leftarrow K-1} \equiv 1$ ($\lambda = 0$). Generic step $k \to k-1$: combine $\mu_{\leftarrow k}$ with the local evidence at k (Lemma 2) to get $\mathcal{N}(m_k^c, v_k^c)$, then invert the transition $a_k = \alpha a_{k-1} + \eta$:

$$\mu_{\leftarrow k-1} = \int \mathcal{N}(a_k; \alpha a_{k-1}, \sigma_\eta^2)\,\mathcal{N}(a_k; m_k^c, v_k^c)\,\mathrm{d}a_k = \mathcal{N}\Big(a_{k-1};\ \frac{m_k^c}{\alpha},\ \frac{v_k^c + \sigma_\eta^2}{\alpha^2}\Big),$$

because viewing the integrand as a function of a_{k-1}, the exponent is quadratic with the stated mean and variance (equivalently: apply Lemma 3 to the inverted relation $a_{k-1} = (a_k - \eta)/\alpha$).

Derivation : combination = RTS smoother

At each node, (25) multiplies three Gaussians (Lemma 2 twice):

$$p(a_k \,|\, x) = \mathcal{N}\big(m_k^{\mathrm{post}}, v_k^{\mathrm{post}}\big), \qquad \frac{1}{v_k^{\mathrm{post}}} = \frac{1}{v_k^{\to}} + \frac{e^{-2t}}{\Delta_t} + \frac{1}{v_k^{\leftarrow}}, \qquad \frac{m_k^{\mathrm{post}}}{v_k^{\mathrm{post}}} = \frac{m_k^{\to}}{v_k^{\to}} + \frac{e^{-t}}{\Delta_t}x_k + \frac{m_k^{\leftarrow}}{v_k^{\leftarrow}}.$$

The forward–backward combination is the *Rauch–Tung–Striebel smoother* [10]. Substituting m_k^{post} into Tweedie (14) gives the k-th score component.

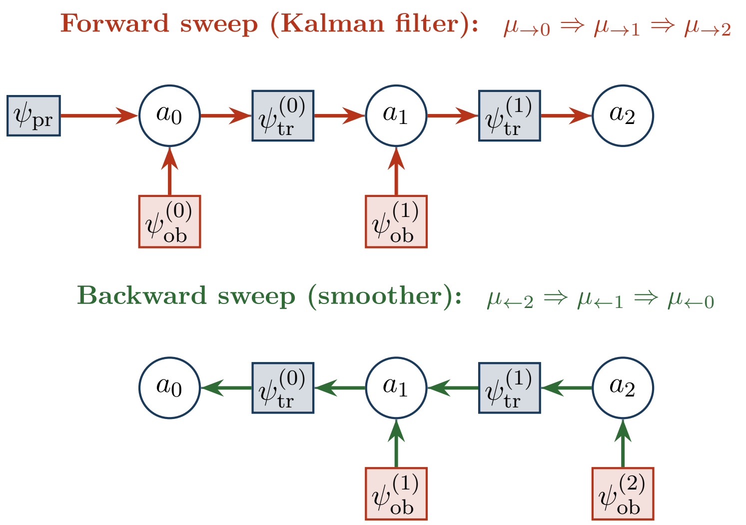

Figure 2: The two sweeps for $K = 3$, drawn on the factor nodes. Forward (red): information from the prior and from x_0, x_1 flows left to right; the forward message into a_k has already absorbed $x_0, \ldots, x_{k-1}$ (and not x_k: Convention A). Backward (green): information from x_2, x_1 flows right to left. The posterior at a_k multiplies forward message, local evidence, and backward message — each observation counted exactly once.

9.3 A complete numerical instance

Fix $\alpha = 0.7$, $t = 0.5$, $x = (1.2,\ -0.4,\ 0.8)$. Then

$$e^{-t} = 0.606531, \quad \Delta_t = 0.632121, \quad \sigma_\eta^2 = 0.51, \quad \tfrac{e^{-2t}}{\Delta_t} = 0.581977, \quad \tfrac{e^{-t}}{\Delta_t} = 0.959517.$$

The pseudo-observations (26) are $\mathcal{N}(e^t x_k,\ \Delta_t e^{2t}) = \mathcal{N}(1.648721\, x_k,\ 1.718282)$. Executing the sweeps:

message	at a_0	at a_1	at a_2
forward $\mu_{\to k}$ (mean, var)	$(0,\ 1)$	$(+0.509486,\ 0.819739)$	$(+0.092348,\ 0.781939)$
evidence (26) (mean, var)	$(+1.978466,\ 1.718282)$	$(-0.659489,\ 1.718282)$	$(+1.318977,\ 1.718282)$
backward $\mu_{\leftarrow k}$ (mean, var)	$(+0.054410,\ 3.585865)$	$(+1.884253,\ 4.547514)$	flat $(\lambda = 0)$
posterior (mean, var)	$(+0.626915,\ 0.537389)$	$(+0.322520,\ 0.494614)$	$(+0.475974,\ 0.537389)$

Sample hand-checks (all reproduced in notebook `02_belief_propagation`):

- *Forward at a_1*: combine $(0, 1)$ with the evidence at node 0: precision $1+0.581977 = 1.581977$, variance 0.632121, mean $0.632121 \times (0 + 0.959517 \times 1.2) = 0.727846$; predict: mean $0.7 \times 0.727846 = 0.509492$, variance $0.49 \times 0.632121 + 0.51 = 0.819739$. (Matches the table to rounding.)
- *Backward at a_1*: combine flat with evidence at node 2: $(1.318977, 1.718282)$; invert the transition: mean $1.318977/0.7 = 1.884253$, variance $(1.718282 + 0.51)/0.49 = 4.547514$. ✓
- *Posterior at a_1*: add the three precisions $1/0.819739 + 0.581977 + 1/4.547514 = 2.021779$, variance 0.494614; the mean follows from the potentials. ✓

The matrix route (17) gives posterior mean $(0.626915, 0.322520, 0.475974)$ and variances $(0.537389,\ 0.494614,\ 0.537389)$ — identical. Tweedie (14) then yields the score

$$S(x, 0.5) = (-1.296836,\ +0.942254,\ -0.808876),$$

which equals $-Q_t x$ from (13) to machine precision. Note the sign of S_1: although $x_1 = -0.4$ is negative, its neighbours pull the posterior mean of a_1 *positive* ($+0.32$), so the score pushes x_1 *up* — a purely joint effect, impossible for the marginal score $-x_1$ to produce… this is Remark 1 in numbers.

10 General K: Gaussian closure and the Kalman connection

For general K nothing changes: the recursions (22)–(24) run for $K-1$ steps each, and the combination (25) holds at every node. Each sweep is a sequence of update (Lemma 2) and predict (Lemma 3) operations on (λ, θ) pairs.

Theorem 13 (Gaussian closure — answering "are the messages Gaussian?")**.** *For the model* (3)–(6)*, every BP message and every belief is exactly Gaussian, at every node, for every K, α, t; consequently the posterior marginals are Gaussian, and BP computes them exactly.*

Derivation 13

By induction along each sweep. The initial messages are Gaussian ((21): the prior; (23): the flat limit $\lambda \to 0$ of a Gaussian family). Each recursion step consists of one product of Gaussians in the same variable (Lemma 2) and one Gaussian propagation (Lemma 3), both of which return Gaussians. The combination (25) is a product of three Gaussians. Exactness then follows from tree-exactness of BP (Proposition 10 and [8, Th. 14.1]). $\square$

Reading : closure is algebra, not a central limit theorem

The closure holds at every node degree — on the degree-2 chain just as on a dense graph — because it is a property of the Gaussian *family* under multiplication and affine marginalisation, not of any aggregation of many weak influences. This resolves the doubt raised in discussion: no CLT is invoked anywhere, so there is no "CLT on two points" to fail *for BP*. The CLT enters only when one tries to justify the *AMP* closure (§12) — and there, on the chain, it does fail, exactly as anticipated.

Reading : the classical identifications

Forward sweep = Kalman filter [5]; forward+backward combination = Rauch–Tung–Striebel smoother [10]; both are the Gaussian specialisation of sum–product on the chain [1, 6]. Tweedie (14) then says: *the k-th joint score component is an affine function of the Kalman-smoothed estimate of frame k* — the joint score of a Markov sequence is a smoothing problem, solved in $O(K)$.

Numerical check : BP reproduces the matrix posterior and the exact score

`numerical_audit.py` sweeps $K \in \{2, 3, 5, 8, 16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1, 3\}$ (80 configurations, random x): maximum disagreement between BP and the matrix route (17) is 1.3×10^{-15} (means), 3.9×10^{-15} (variances), 1.2×10^{-14} (scores) — double-precision rounding accumulated along the recursion. BP *is* the score, computed through the precision matrix in $O(K)$.

11 The chain solved in closed form: bulk analysis

So far BP is an *algorithm*; in this section we solve its fixed point in closed form in the *bulk* of the chain — the homogeneous interior of a long chain, far from both boundaries — and obtain explicit formulas for the posterior variance, the full posterior covariance, and the correlation length of the score. These formulas are the backbone of the BP/AMP comparison (§12) and of Experiment A (§13).

11.1 Bulk parameters

In the interior, the posterior precision (17) is the tridiagonal Toeplitz matrix with

$$\boxed{J_{\mathrm{d}} = \frac{e^{-2t}}{\Delta_t} + \frac{1+\alpha^2}{\sigma_\eta^2}, \qquad \beta = -\frac{\alpha}{\sigma_\eta^2}} \qquad (\sigma_\eta^2 = 1-\alpha^2). \tag{27}$$

J_{d} has two parts with opposite behaviour in t: the *evidence* precision e^{-2t}/Δ_t (the tilted-measure pinning strength, divergent as $1/2t$ at small t, vanishing at large t) and the *prior* precision $(1+\alpha^2)/\sigma_\eta^2$ (constant). The coupling β is pure prior. The single dimensionless ratio $|\beta|/J_{\mathrm{d}} \in (0, \frac{1}{2})$ controls everything below.

11.2 The BP fixed point and the exact bulk variance

Write Gaussian BP directly in the (J, h) parametrisation: the message from node k to node $k{+}1$ is Gaussian with some precision, and on the homogeneous interior the precision λ_k of the *one-sided summary* (everything to the left of $k{+}1$, evidence included) obeys

$$\lambda_{k+1} = J_{\mathrm{d}} - \frac{\beta^2}{\lambda_k}, \tag{28}$$

obtained by integrating a_k out of $\exp(-\frac{1}{2}\lambda_k a_k^2 + \theta_k a_k - \beta a_k a_{k+1} - \dots)$: the Gaussian integral contributes $-\beta^2/\lambda_k$ to the precision seen by a_{k+1}. (This is the same algebra as §9 in a different parametrisation; the audit checks the *results* agree.)

Theorem 14 (BP bulk fixed point — always exists)**.** *The recursion* (28) *has the fixed points* $\lambda^\pm = (J_{\mathrm{d}} \pm \sqrt{J_{\mathrm{d}}^2 - 4\beta^2})/2$. *For every* $\alpha \in (-1, 1)$ *and every* $t > 0$*,*

$$J_{\mathrm{d}} - 2|\beta| \;=\; \frac{e^{-2t}}{\Delta_t} + \frac{(1-|\alpha|)^2}{\sigma_\eta^2} \;>\; 0,$$

so the discriminant is strictly positive, the upper fixed point

$$\boxed{\lambda^* = \frac{J_{\mathrm{d}} + \sqrt{J_{\mathrm{d}}^2 - 4\beta^2}}{2}} \tag{29}$$

exists, and it is strictly attractive ($|f'(\lambda^*)| = \beta^2/\lambda^{*2} < 1$)*. Recombining the two directions and the local term,*

$$\Lambda_{\mathrm{bulk}} = J_{\mathrm{d}} - \frac{\beta^2}{\lambda^*} - \frac{\beta^2}{\lambda^*} = \sqrt{J_{\mathrm{d}}^2 - 4\beta^2} \qquad \Longrightarrow \qquad \boxed{V_{\mathrm{exact}} = \frac{1}{\sqrt{J_{\mathrm{d}}^2 - 4\beta^2}}} \tag{30}$$

Derivation 14

Existence. $1+\alpha^2-2|\alpha| = (1-|\alpha|)^2 \geq 0$ (AM–GM, with equality only at $|\alpha|=1$, excluded), and the evidence term is positive for every finite t; hence $J_{\mathrm{d}} > 2|\beta|$ strictly and the square root is real. **Stability.** The map $f(\lambda) = J_{\mathrm{d}} - \beta^2/\lambda$ has $f'(\lambda) = \beta^2/\lambda^2$; since $\lambda^* \geq J_{\mathrm{d}}/2 > |\beta|$, we get $f'(\lambda^*) < 1$: the forward sweep converges geometrically to λ^* from any positive initialisation, and the boundary effects of a finite chain decay at the same geometric rate. **Recombination.** The marginal precision at a bulk node is the local J_{d} plus the two one-sided corrections $-\beta^2/\lambda^*$. Using $\lambda^{*2} - J_{\mathrm{d}}\lambda^* + \beta^2 = 0$, i.e. $\beta^2/\lambda^* = J_{\mathrm{d}} - \lambda^*$: $\Lambda_{\text{bulk}} = J_{\mathrm{d}} - 2(J_{\mathrm{d}} - \lambda^*) = 2\lambda^* - J_{\mathrm{d}} = \sqrt{J_{\mathrm{d}}^2 - 4\beta^2}$. □

11.3 The full bulk covariance and the correlation length of the score

Theorem 15 (Geometric posterior covariance)**.** *In the bulk, the posterior covariance decays geometrically with frame distance:*

$$\boxed{(J^{-1})_{i,\,i+d} = \frac{q^{\,d}}{\sqrt{J_{\mathrm{d}}^2 - 4\beta^2}}, \qquad q = \frac{J_{\mathrm{d}} - \sqrt{J_{\mathrm{d}}^2 - 4\beta^2}}{2\,|\beta|} \in (0,1).} \tag{31}$$

Derivation 15: transfer-matrix ansatz

For an infinite tridiagonal Toeplitz J, seek the row of the inverse in the form $(J^{-1})_{i,i+d} = C\,q^{\,d}$ for $d \geq 0$ (symmetric in d). The defining relation $\sum_j J_{ij}(J^{-1})_{j,i+d} = \delta_{0d}$ reads, for $d \geq 1$,

$$\beta\,Cq^{\,d-1} + J_{\mathrm{d}}\,Cq^{\,d} + \beta\,Cq^{\,d+1} = 0 \quad\Longleftrightarrow\quad \beta q^2 + J_{\mathrm{d}} q + \beta = 0,$$

whose root inside the unit disc is $q = (J_{\mathrm{d}} - \sqrt{J_{\mathrm{d}}^2 - 4\beta^2})/(2|\beta|)$ (for $\beta < 0$, i.e. $\alpha > 0$, the inverse entries are all positive; for $\alpha < 0$ they alternate in sign, $q \mapsto -q$, with the same modulus). The $d=0$ relation fixes C: $J_{\mathrm{d}}C + 2\beta Cq = 1$ and $2\beta q = -(J_{\mathrm{d}} - \sqrt{J_{\mathrm{d}}^2 - 4\beta^2})$ give $C = 1/\sqrt{J_{\mathrm{d}}^2 - 4\beta^2} = V_{\text{exact}}$, consistently with (30). □

Reading : q is the memory of the posterior — with the right limits

q measures how far evidence travels along the chain: the posterior correlation length is $\xi = 1/\log(1/q)$ frames. Its limits are exactly what they must be:

$$t \to 0: \quad q \approx \frac{|\beta|}{J_{\mathrm{d}}} \to 0 \quad \text{(each frame pinned by its own data: near-diagonal posterior);}$$

$$t \to \infty: \quad q \to \frac{(1+\alpha^2) - (1-\alpha^2)}{2|\alpha|} = |\alpha| \quad \text{(no data: the posterior \textit{is} the prior).}$$

So as diffusion time runs, the score's internal correlation length climbs monotonically from 0 to the prior's own $1/\log(1/|\alpha|)$. This single number $q(\alpha, t)$ will reappear as the *exact decay rate of the locality error* in Experiment A.

Numerical check : bulk formulas vs brute force

At the centre of a $K = 400$ chain, over 20 configurations ($\alpha \in \{0.2, 0.5, 0.8, 0.95, -0.6\}$, $t \in \{0.05, 0.3, 1, 3\}$): V_{exact} matches $\mathrm{diag}(J^{-1})$ to relative error 2.2×10^{-11}; the geometric law (31) matches $(J^{-1})_{i,i+d}$ for $d \leq 5$ to 8.1×10^{-13}; the recombination identity in (30) holds to 1.4×10^{-14} on a 2010-point (α, t) grid; and the existence margin $J_{\mathrm{d}} - 2|\beta|$ is positive over the entire grid.

12 Approximate message passing: same score, different variance, explicit breakdown

12.1 The general idea, and why our model is its opposite

AMP — in the physics literature, the TAP equations — is the message passing one uses when exact messages cannot be tracked and the factor graph is *dense* [2, 8, 13]. For a linear model $y = \Phi a + w$ with dense $\Phi \in \mathbb{R}^{m \times N}$, every factor touches all N variables; each message is then a sum of many weakly dependent contributions, hence *approximately Gaussian by a central limit argument*, and one propagates only means and variances, with the Onsager reaction term subtracting each node's back-influence. Rigorously, Genovese and Piana [4] show for ℓ_2 minimisation that the BP means satisfy the AMP equations asymptotically as $N \to \infty$ — and note that on a *tree* BP is already exact, so AMP has nothing to add there.

Our model is the opposite of this regime in every respect: the measurement operator is *diagonal* ($x = e^{-t}a + \text{noise}$), the coupling lives in the *prior*, and every node has at most two neighbours. To make the comparison concrete we view the posterior (17) as a Gaussian Markov random field with precision J and field h and ask three solvers for its marginals:

$$\text{mean field: } V_i = \frac{1}{J_{ii}}; \qquad \text{AMP/TAP: } V_i = \frac{1}{J_{ii} - \sum_{k \neq i} J_{ik}^2 V_k}; \qquad \text{BP/exact: } V_i = (J^{-1})_{ii}.$$

The AMP/TAP closure is the cavity variance with the "exclude one neighbour" correction dropped ($V_{k \to i} \approx V_k$): harmless when each node has many weak neighbours (removing one is an $O(1/N)$ change — the dense regime), an $O(1)$ modification on a chain where each node has two.

12.2 The score is closure-independent

Theorem 16 (Same mean ⇒ same score)**.** *For a Gaussian posterior, the stationarity condition for the mean is the same linear system $Jm = h$ under mean field, BP, and AMP. Hence all three return the exact posterior mean, and — via Tweedie (14) — the same, exact, joint score, for every (K, α, t), regardless of whether the variance closure is accurate or even solvable.*

Derivation 16

In each scheme the update for the mean of node i has the form $m_i \leftarrow (h_i - \sum_{k \neq i} J_{ik}\, m_k)/J_{ii}$ (with scheme-dependent corrections that vanish at the fixed point: the cavity exclusion affects which V's appear, not the linear relation among the means). At any fixed point, $J_{ii} m_i + \sum_{k \neq i} J_{ik} m_k = h_i$ for all i, i.e. $Jm = h$, whose unique solution is the exact posterior mean. The variance closure modifies only the *uncertainty* bookkeeping. Tweedie converts the mean into the score; the score is therefore closure-independent. □

Reading : why this is the precise answer to "is AMP = BP here?"

For the *score* — the object diffusion actually needs — yes: BP, AMP and even naive mean field coincide, because the score only needs the posterior mean and the mean solves a closure-independent linear system. For the *posterior variances* — the object that message-passing theory needs internally (and that controls error bars, likelihoods, and any nonlinear extension) — no: the closures differ, measurably and, as we now show, catastrophically at strong coupling.

12.3 The AMP bulk fixed point, its breakdown time, and the critical coupling

In the bulk, the AMP closure reads $V = 1/(J_\mathrm{d} - 2\beta^2 V)$ — compare the BP cavity $V_\mathrm{cav} = 1/(J_\mathrm{d} - \beta^2 V_\mathrm{cav})$ of (28): *the entire difference between BP and AMP on the chain is the factor 2*, both neighbours included where BP excludes one.

Theorem 17 (AMP bulk variance and existence)**.** *The AMP closure is the quadratic* $2\beta^2V^2 - J_\mathrm{d}V + 1 = 0$*, with physical root*

$$\boxed{V_{AMP} = \frac{J_\mathrm{d} - \sqrt{J_\mathrm{d}^2 - 8\beta^2}}{4\beta^2}} \tag{32}$$

(the branch continuously connected to the decoupled limit $1/J_\mathrm{d}$ *as* $\beta \to 0$*), which exists iff*

$$\boxed{J_\mathrm{d} \;\ge\; 2\sqrt{2}\,|\beta|} \tag{33}$$

— a strictly stronger requirement than BP's $J_\mathrm{d} \ge 2|\beta|$ *(Theorem 14), which always holds. Solving (33) as an equality for* t*, with* $u := e^{-2t}$ *and* $e^{-2t}/\Delta_t = u/(1-u)$*:*

$$t_c(\alpha) = -\tfrac{1}{2}\log\frac{g}{1+g}, \qquad g(\alpha) = \frac{2\sqrt{2}\,|\alpha| - 1 - \alpha^2}{1-\alpha^2}, \tag{34}$$

finite exactly when $g > 0$*, i.e.*

$$\boxed{|\alpha| > \alpha_c = \sqrt{2} - 1 \approx 0.4142} \tag{35}$$

For $|\alpha| \le \alpha_c$ *the AMP variance has a fixed point at every diffusion time; for* $|\alpha| > \alpha_c$ *it ceases to exist for all* $t > t_c(\alpha)$*.*

Derivation 17

Quadratic and branch. Multiply $V(J_\mathrm{d} - 2\beta^2V) = 1$ out; the roots are $V_\pm = (J_\mathrm{d} \pm \sqrt{J_\mathrm{d}^2 - 8\beta^2})/(4\beta^2)$. As $\beta \to 0$, $V_- \to 1/J_\mathrm{d}$ (expand the square root) while V_+ diverges; moreover V_- is the attractive fixed point of the damped iteration used in practice. Real roots require $J_\mathrm{d}^2 \ge 8\beta^2$. **Breakdown time.** The map $t \mapsto J_\mathrm{d}(t)$ is strictly decreasing (the evidence dies as t grows) with range $(\,(1+\alpha^2)/\sigma_\eta^2,\ \infty)$, so the equality $J_\mathrm{d} = 2\sqrt{2}|\beta|$ has at most one solution. Substituting $u/(1-u)$ for the evidence term and solving the linear equation in u gives (34). **Critical coupling.** $g > 0 \iff \alpha^2 - 2\sqrt{2}|\alpha| + 1 < 0 \iff |\alpha| \in (\sqrt{2}-1, \sqrt{2}+1)$; with $|\alpha| < 1$ this is $|\alpha| > \sqrt{2}-1$. Equivalently: at $t = \infty$ the posterior is the bare prior, $J \to Q_0$, and (33) for Q_0 itself reads $(1+\alpha^2) \ge 2\sqrt{2}|\alpha|$ — AMP applied to a bare AR(1) chain works iff $|\alpha| \le \sqrt{2}-1$. ☐

Theorem 18 (Weak-coupling law: AMP errs at fourth order, upward)**.** *Expanding* (30) *and* (32) *in* $\epsilon := \beta^2/J_d^2$,

$$V_{\text{exact}} = \frac{1}{J_d}\big(1 + 2\epsilon + 6\epsilon^2 + O(\epsilon^3)\big), \qquad V_{\text{AMP}} = \frac{1}{J_d}\big(1 + 2\epsilon + 8\epsilon^2 + O(\epsilon^3)\big),$$

so the closures agree to second order in the coupling and

$$\boxed{V_{\text{AMP}} - V_{\text{exact}} = \frac{2\beta^4}{J_d^5} + O\Big(\frac{\beta^6}{J_d^7}\Big)} \qquad \text{— AMP always overestimates the bulk variance.} \tag{36}$$

Derivation 18

$(1-4\epsilon)^{-1/2} = 1+2\epsilon+6\epsilon^2+O(\epsilon^3)$ gives the exact expansion. For AMP, $1-\sqrt{1-8\epsilon} = 4\epsilon + 8\epsilon^2 + O(\epsilon^3)$, and multiplying by $J_d/(4\beta^2) = 1/(4\epsilon J_d)$ gives the stated series. Subtracting, $V_{\text{AMP}} - V_{\text{exact}} = 2\epsilon^2/J_d = 2\beta^4/J_d^5$ at leading order. □

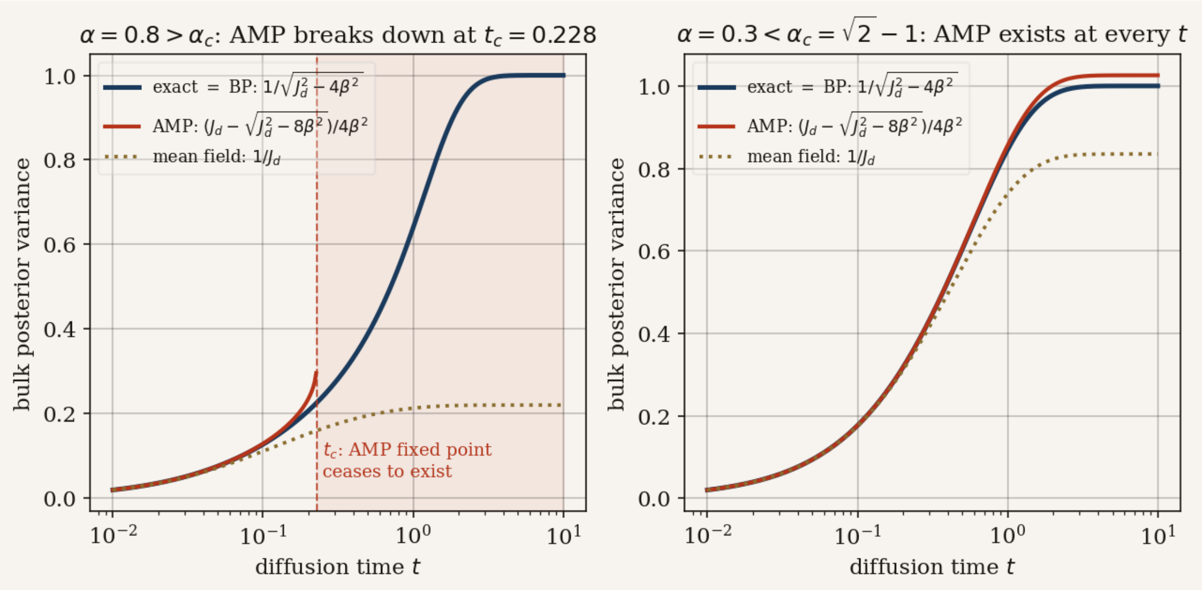

Figure 3: **Bulk posterior variance: the three closures in closed form.** Left ($\alpha = 0.8 > \alpha_c$): exact = BP (navy), AMP (32) (red), mean field (gold). The AMP branch terminates with the characteristic square-root singularity at the exact breakdown time $t_c(0.8) = 0.228$ from (34) (shaded: no fixed point). Right ($\alpha = 0.3 < \alpha_c$): AMP exists at every t and slightly overestimates the variance at large t, as predicted by (36).

Reading : the precise content of "a CLT on two points"

J. Garnier-Brun's remark — that AMP's Gaussian-message assumption is a central limit statement that "on two data points is usually not right" — is here made exact. On the chain, dropping the cavity exclusion replaces β^2 by $2\beta^2$ in the fixed-point equation; the discriminant moves from $J_d^2 - 4\beta^2$ (never negative) to $J_d^2 - 8\beta^2$ (negative beyond (34)). At weak coupling the sin is venial ((36): a fourth-order overestimate); at strong coupling it is mortal (no fixed point at all). Meanwhile on a dense problem the same closure is excellent — the audit's dense gate ($N = 600$ random SPD precision) shows AMP variance error 1.7×10^{-3} against mean field's 4×10^{-2}. AMP is not wrong; it is *built for a different*

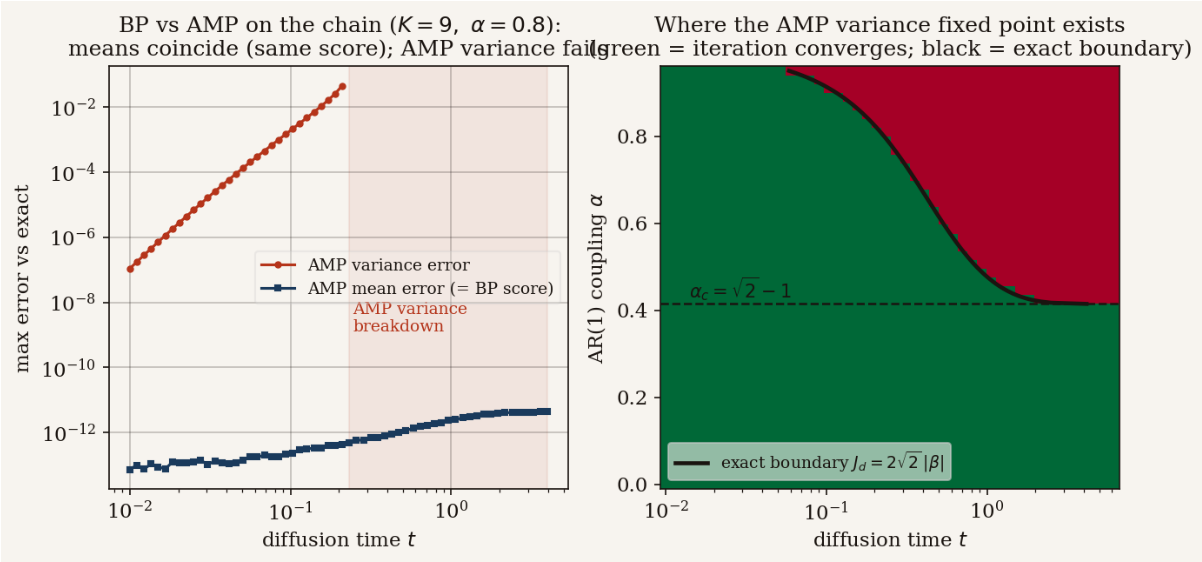

Figure 4: **BP vs. AMP on a finite chain ($K = 9$, $\alpha = 0.8$).** Left: the AMP *mean* error (navy, $\sim 10^{-12}$, flat in t) — the score is closure-independent (Theorem 16); the AMP *variance* error (red) grows with t until the closure stops converging (shaded). Right: where the AMP variance fixed point exists in the (α, t) plane; green = the self-consistent iteration converges, and the black curve is the *exact* boundary $t_c(\alpha)$ of (34) with the horizontal dashed line at $\alpha_c = \sqrt{2} - 1$. Theory and iteration agree at all 180 scanned points.

graph.

Numerical check : every claim of this section, measured

(i) AMP mean = exact mean to 4.3×10^{-12} and AMP score = BP score to 1.7×10^{-12} over 36 chain configurations (Theorem 16). (ii) V_{AMP} from (32) matches the converged self-consistent iteration at the centre of a $K = 400$ chain to 2.5×10^{-12} (12 configurations). (iii) The existence condition (33) predicts the convergence / non-convergence of the iteration at all 180 points of an (α, t) scan. (iv) t_c from (34) matches the bisected flip point of the iteration to 6.6×10^{-6} relative. (v) Below α_c the iteration converges even at $t = 50$. (vi) The weak-coupling ratio $(V_{\mathrm{AMP}} - V_{\mathrm{exact}})/(2\beta^4/J_{\mathrm{d}}^5) \to 1$ within 0.4% at the smallest couplings tested.

13 Experiment A: local messages versus full sweeps

How much does a frame's score depend on *distant* frames? This was one of the two experimental questions posed in discussion, and the bulk analysis of §11 lets us answer it exactly rather than only empirically.

13.1 The local estimator and the exact error measure

Define the *local estimator of radius* r as the posterior mean of a_k given only the observations within distance r,

$$m_k^{(r)} := \mathbb{E}\big[a_k \,\big|\, x_{k-r}, \dots, x_{k+r}\big],$$

the rest of the chain marginalised out, and the corresponding score via Tweedie. Two structural facts make this a *clean* experiment:

- **Exactness on the window.** Any contiguous block of a stationary AR(1) chain is itself a stationary AR(1) chain, so $m_k^{(r)}$ is computed *exactly* on the windowed sub-chain — the only approximation in play is the truncation of range. At $r \geq K-1$ the window is the whole chain and the estimator is exact (audited: 3.6×10^{-15}).
- **A deterministic error.** Both m_k and $m_k^{(r)}$ are linear in x, so their difference is $e^\top x$ for a computable row vector e, and the root-mean-square error over the data distribution $x \sim P_t = \mathcal{N}(0, \Sigma_t)$ is the closed-form number

$$\mathrm{RMS}_r(k) = \sqrt{e^\top \Sigma_t\, e}$$

— no sampling, no seed, nothing to tune. (Code: `local_bp.rms_truncation_error`.)

13.2 The locality law

Theorem 19 (Locality error decays exactly as q^r)**.** *In the bulk,* $\log \mathrm{RMS}_r(k)$ *decreases linearly in* r *with slope* $\log q$*, where* $q = q(\alpha, t)$ *is the posterior correlation decay rate of Theorem 15. Equivalently: truncating the message range at* r *costs a factor* q *per discarded frame.*

Derivation 19: why the rate must be q — and how it is verified

The error of the radius-r estimator is the influence on $\mathbb{E}[a_k \mid \cdot\,]$ of the evidence beyond the window. In the Gaussian chain, the influence of evidence at frame $k \pm d$ on the posterior mean at k is proportional to the posterior covariance $(J^{-1})_{k,k\pm d}$, which by Theorem 15 decays as q^d. The leading discarded contribution sits at distance $r+1$, so the error inherits the geometric rate q per unit radius (the prefactor sums the geometric tail and the data covariance). Rather than chase the prefactor analytically, we verify the *rate*: the audit fits the slope of $\log \mathrm{RMS}_r$ over $r = 2, \ldots, 13$ at the centre of a $K = 121$ chain and finds it equal to $\log q$ within 0.2% at $t \in \{0.1, 0.5, 2.0\}$ $(\alpha = 0.8)$.

Reading : the answer to "local messages vs full sweeps"

Strictly-neighbour messages ($r = 1$) are exact at $t = 0$, excellent at small t, and wrong by a factor governed by q^2 relative reach at intermediate t; the full sweeps are exact always. The cost of locality is not a constant — it is a *rate*, $q(\alpha, t)$ per frame of range, and that rate is largest precisely in the mid-diffusion regime where generative sampling spends its critical steps (cf. the tilted-measure crossover of §5). In the Gaussian case the local approximation is thus completely *quantified*: choose the radius r as $r \gtrsim \xi \log(1/\varepsilon)$ with $\xi = 1/\log(1/q)$ to guarantee relative error ε.

14 Experiment B: lifecycle of the precision matrix

The score (13) is entirely controlled by $Q_t = \Sigma_t^{-1}$, and a zero of Q_t is a conditional independence of P_t. This experiment tracks how the sparsity pattern of Q_t is born, lives and dies along diffusion time — the question raised in the project notebook ("the zeros should progressively fill up… then identity again").

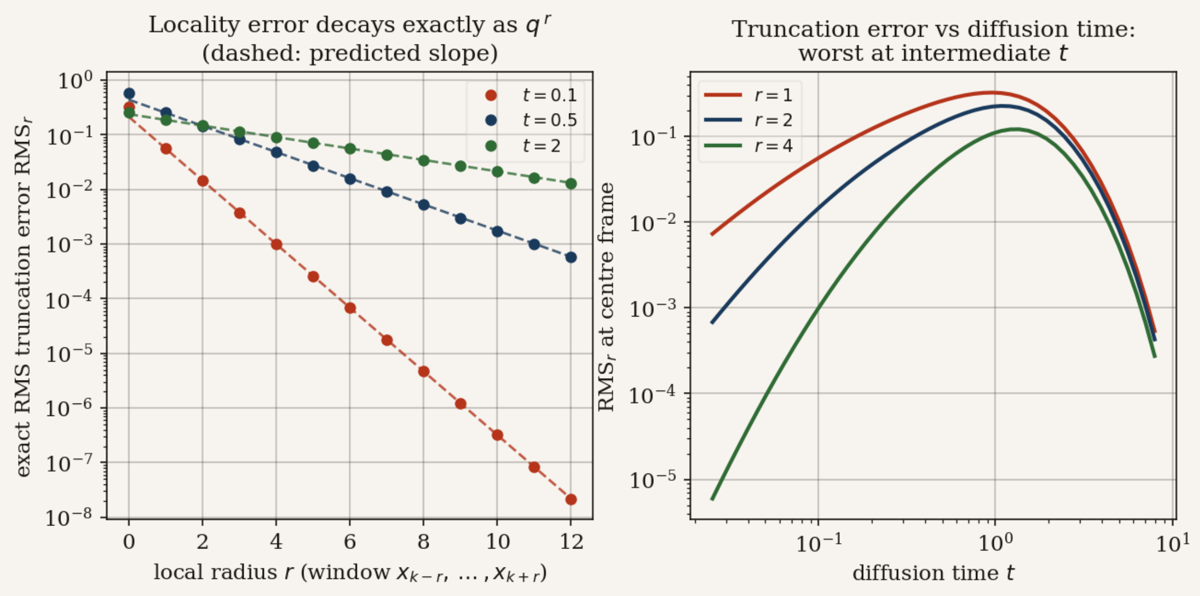

Figure 5: **Locality of the joint score.** Left: exact RMS truncation error of the radius-r estimator (dots) at three diffusion times; the dashed lines have the *predicted* slopes $q(\alpha,t)^r$ from (31) — no fitting. Right: the same error at fixed radius as a function of t: it vanishes at $t \to 0$ (each frame pinned by its own observation: $q \to 0$) and at $t \to \infty$ (the *score* decouples as $Q_t \to I$, and the signal itself dies), and peaks at intermediate t.

14.1 Three equivalent forms of Q_t

Theorem 20 (Three forms)**.** *For all $t \geq 0$, with $\gamma_t := e^{-2t}/\Delta_t$ (the SNR) and $c_t := e^{2t} - 1$:*

$$Q_t = \tfrac{1}{\Delta_t}\big(I + \gamma_t\,\Sigma_0\big)^{-1} \qquad \textit{(SNR form)}, \tag{37}$$

$$Q_t = U\ \mathrm{diag}\Big(\frac{1}{e^{-2t}\omega_i + \Delta_t}\Big)\,U^\top, \qquad \Sigma_0 = U\,\mathrm{diag}(\omega_i)U^\top \qquad \textit{(spectral form)}, \tag{38}$$

$$Q_t = e^{2t}\,Q_0\big(I + c_t Q_0\big)^{-1} = e^{2t}\sum_{m\geq 0}(-c_t)^m\,Q_0^{m+1} \qquad \textit{(resolvent form)}. \tag{39}$$

Derivation 20

SNR. Factor Δ_t out of (12) and invert. **Spectral.** I commutes with everything: $\Sigma_t = U\,\mathrm{diag}(e^{-2t}\omega_i + \Delta_t)U^\top$, and inverting a diagonal is entry-wise. **Resolvent.** Multiply (12) on the left by Q_0: $Q_0\Sigma_t = e^{-2t}(I + c_t Q_0)$ using $\Delta_t e^{2t} = c_t$; invert both sides. The Neumann series converges for $\|c_t Q_0\| < 1$, the small-t / high-SNR regime. □

Reading : locality is a basis-dependent statement

The spectral form says the eigenvectors of Σ_t and Q_t are those of Σ_0 *for every* t — only the eigenvalues flow, $\omega_i \mapsto e^{-2t}\omega_i + \Delta_t \to 1$. In the Σ_0-eigenbasis (near-Fourier modes, since Σ_0 is symmetric Toeplitz) the denoising problem decouples into K scalar problems at all times: maximal locality. In the *frame* basis — the operationally meaningful one for video — Q_t is tridiagonal only at $t = 0$. Both statements describe the same matrix; sparsity is a property of a matrix *in a basis*.

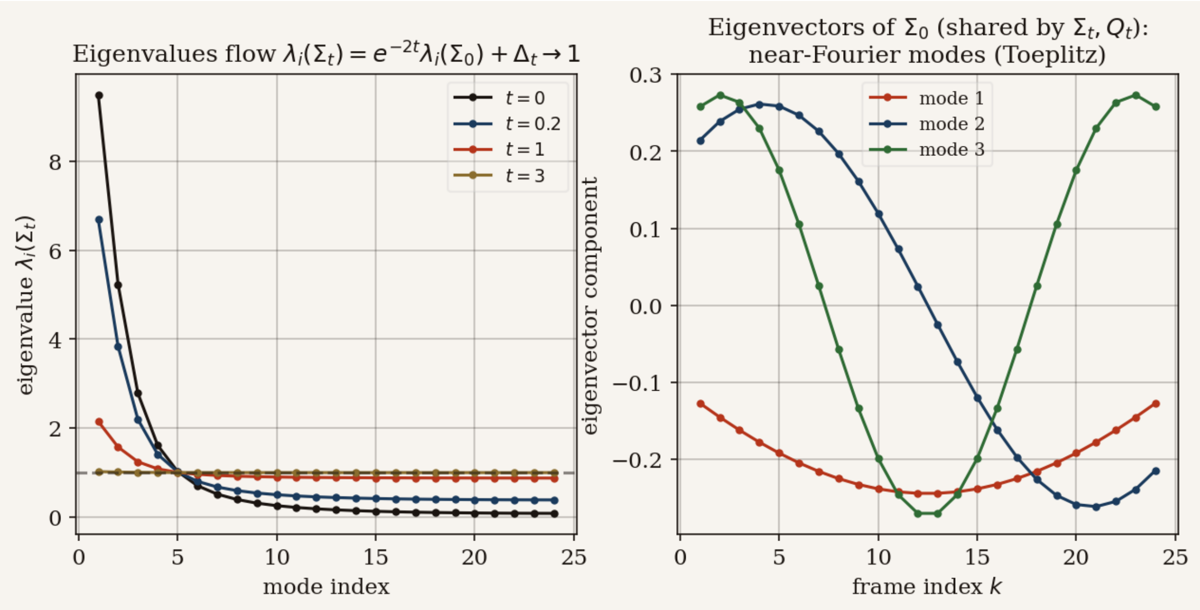

Figure 6: **Spectral lifecycle.** Left: eigenvalues of Σ_t flow as $e^{-2t}\omega_i + \Delta_t$, all toward 1. Right: the shared eigenvectors of Σ_0 (hence of Σ_t, Q_t at every t) are near-Fourier modes.

14.2 The band-fill law (corrected)

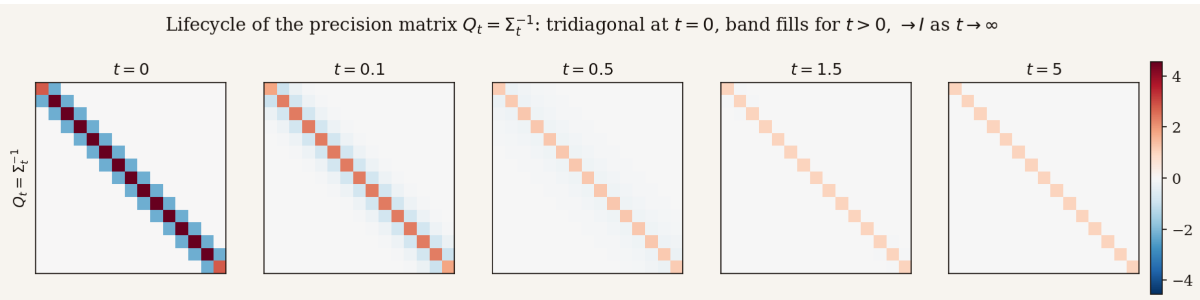

Figure 7: **Lifecycle of Q_t in the frame basis** ($K = 15$, $\alpha = 0.8$). Tridiagonal at $t = 0$ (Markov); the band fills for $t > 0$ (dense); the magnitude collapses and $Q_t \to I$ as $t \to \infty$ (isotropic noise wins).

Theorem 21 (Band-fill leading order — no factorial)**.** *For small t and frame distance $d \geq 1$,*

$$\boxed{(Q_t)_{i,i+d} = (-1)^{d-1}\,(2t)^{d-1}\,(Q_0^d)_{i,i+d} \;+\; O(t^d)} \tag{40}$$

Derivation 21: Neumann series of the resolvent

In (39), substitute $c_t = 2t + 2t^2 + O(t^3)$ and $e^{2t} = 1 + O(t)$:

$$Q_t = (1 + O(t))\big[\,Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \cdots\big].$$

Q_0 is tridiagonal, so Q_0^m has bandwidth m: the entry at distance d receives its first contribution from the term $m = d$, which enters the series with prefactor $(-c_t)^{d-1} = (-1)^{d-1}(2t)^{d-1}(1 + O(t))$. All higher terms are $O(t^d)$. □

Remark 3 (Correction to a previously circulated formula)**.** An earlier internal note (`precision_lifecycle_summary.pdf`, eq. (12)) printed (40) with an extraneous $1/(d-1)!$ factor. That factor does not follow from the resolvent expansion — the leading distance-d term is the *single* contribution $m = d$, with no combinatorial sum behind it — and the audit rejects it: the no-factorial coefficient matches the measured $(Q_t)_{i,i+d}/t^{d-1}$ to better than 0.9% for every $d \in \{1, \dots, 5\}$ ($\alpha = 0.9$, $K = 25$), while the factorial version fails by 86% at $d = 3$, 290% at $d = 4$, 800% at $d = 5$. (Empirical *slope* fits cannot distinguish the two — the slopes $d-1$ are prefactor-blind — which is how the error survived until the coefficient itself was audited. A useful lesson about checking coefficients, not just exponents.)

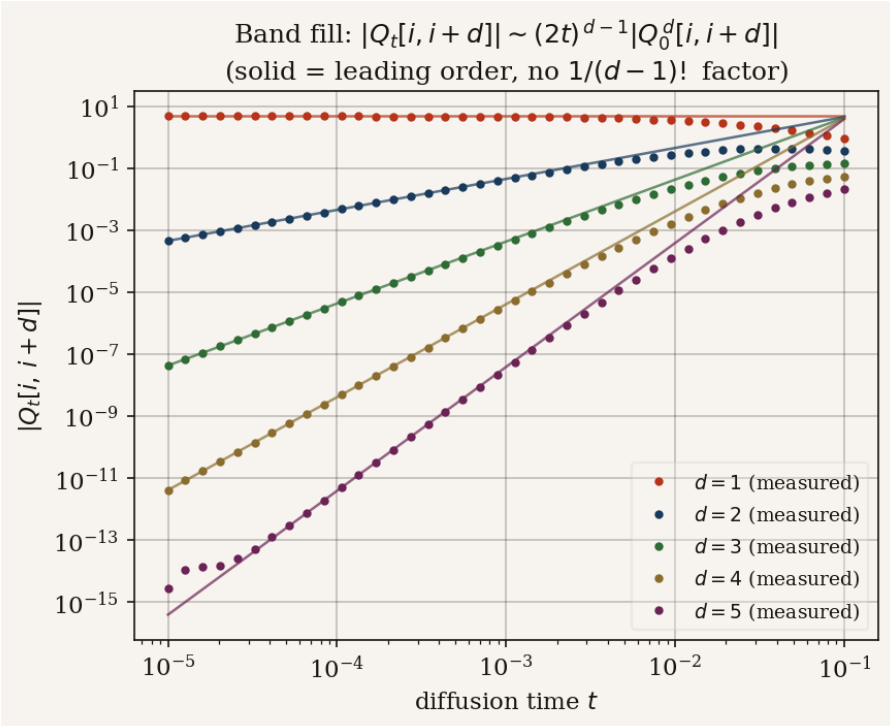

Figure 8: **Band fill, log–log.** Measured $|(Q_t)_{i,i+d}|$ (dots) against the leading order (40) (solid; no factorial). Slopes $d - 1 = 0, \dots, 4$; the agreement at small t is at the percent level for every d.

14.3 Return to the identity, and one number for the whole lifecycle

Proposition 22 (Large-t return)**.** *As $t \to \infty$, writing $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$,*

$$Q_t = I - e^{-2t}(\Sigma_0 - I) + e^{-4t}(\Sigma_0 - I)^2 + O(e^{-6t}),$$

so $\|Q_t - I\|_F \simeq e^{-2t}\|\Sigma_0 - I\|_F$: every correlation dies at the universal rate e^{-2t} and the score collapses to the isotropic $-x$. (Audited at $t \in \{3, 6, 10\}$.)

Reading : Experiments A and B are the same fact, twice

The lifecycle of Q_t (Exp. B) and the locality error (Exp. A) are two views of one object: how far apart frames talk to each other through the score. At $t = 0$ the conversation is nearest-neighbour (Q_t tridiagonal; $q = 0$); at intermediate t it has range $\xi = 1/\log(1/q)$ (band filled; locality error maximal); at $t \to \infty$ it ceases ($Q_t \to I$; everything decouples). The bulk parameter $q(\alpha, t)$ of §11 is the single dial behind both pictures.

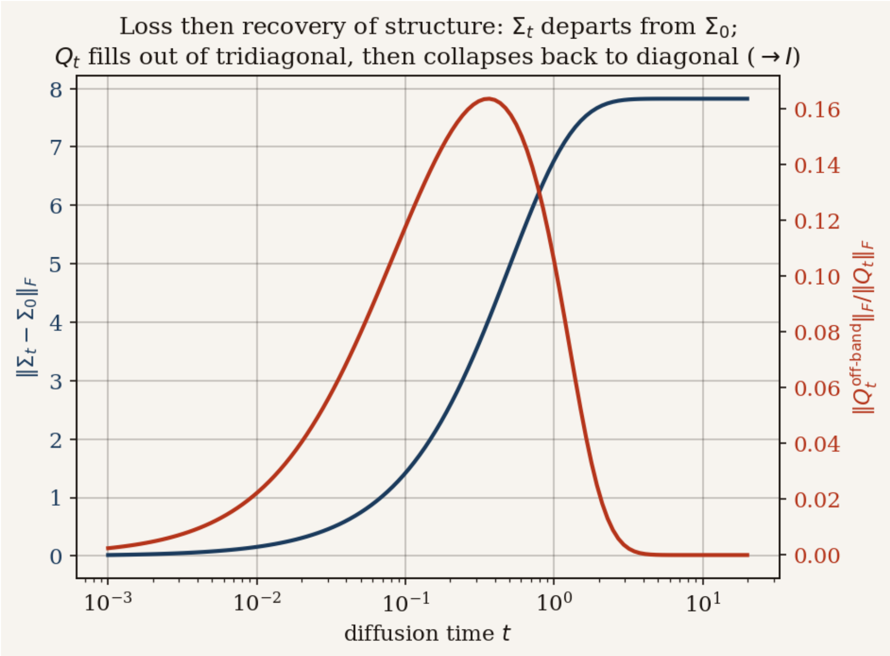

Figure 9: **Loss then recovery of structure, in one curve each.** $\|\Sigma_t - \Sigma_0\|_F$ (navy) grows monotonically as the covariance is driven to I; the off-tridiagonal Frobenius mass of Q_t (red) rises from zero, peaks at intermediate t, and decays back to zero — the band fills, then empties.

15 Summary of established results, and what comes next

15.1 What is established

Every row below is closed-form, derived in this note, and verified by the named audit section (Appendix A).

Result	Where	Audit
Clean precision Q_0 tridiagonal, entries explicit	Prop. 6	§1
$\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$; score $S = -Q_t x$	Prop. 7	§2
Tweedie: $S_k = (e^{-t}\mathbb{E}[a_k\|x] - x_k)/\Delta_t$, two routes agree	Th. 8	§6
$K = 2$ fully explicit; coupling $r = \alpha e^{-2t}$	§6	§10
Posterior $J = (e^{-2t}/\Delta_t)I + Q_0$, $h = (e^{-t}/\Delta_t)x$	Prop. 9	§5,6
Factor graph is a tree; BP exact, $O(K)$; Convention A	Props. 10,12	§5
All messages Gaussian by algebraic closure (no CLT)	Th. 13	§5
BP = Kalman filter + RTS smoother; machine precision	§10	§5
Bulk: $\lambda^* = (J_\mathrm{d}+\sqrt{J_\mathrm{d}^2-4\beta^2})/2$ exists always	Th. 14	§11
Bulk variance $1/\sqrt{J_\mathrm{d}^2 - 4\beta^2}$; covariance $q^d V$	Ths. 14,15	§11
Same mean $\Rightarrow$ BP, AMP, MF give the *same score*	Th. 16	§7
AMP variance $V = (J_\mathrm{d}-\sqrt{J_\mathrm{d}^2-8\beta^2})/4\beta^2$; exists iff $J_\mathrm{d} \geq 2\sqrt{2}\|\beta\|$	Th. 17	§8,11
Exact breakdown time $t_c(\alpha)$; critical coupling $\alpha_c = \sqrt{2} - 1$	Th. 17	§11
Weak coupling: $V_{\text{AMP}} - V_{\text{ex}} = 2\beta^4/J_\mathrm{d}^5 + \dots$	Th. 18	§11
Locality error decays exactly as q^r	Th. 19	§9,11
Band fill $(Q_t)_{i,i+d} = (-1)^{d-1}(2t)^{d-1}(Q_0^d)_{i,i+d} + O(t^d)$, *no* $1/(d-1)!$	Th. 21	§3
Spectral preservation; large-t return $Q_t \to I$ at rate e^{-2t}	Th. 20, Prop. 22	§2,4

15.2 The structural moral

The posterior of a Markov sequence under frame-wise noise *is* a caterpillar tree, and the joint score *is* a smoothing problem on that tree. An architecture that hard-codes this graph gets exactness and $O(K)$ cost for free in the Gaussian case — and retains the correct *structure* (the recursions (22)–(24)) for any Markov prior, Gaussian or not; only the message *parametrisation* changes. This is the same shift that took vision from generic MLPs to convolutions: encode what is known about the data — here, its internal Markov time — into the computation, instead of spending capacity rediscovering it. AMP is the wrong inductive bias for this graph, and §12 now says so with an exact phase boundary rather than a heuristic.

15.3 Research directions

1. **Non-Gaussian prior (Laplace ar(1)).** Gaussian closure is the only Gaussian-specific ingredient of the whole construction. With Laplace innovations the recursions (22)–(24) survive but messages become genuine functions; preliminary work shows the *boundary* message retains a clean closed form while interior messages do not — the natural next document mirrors this one with messages tracked numerically on a grid, the audit discipline unchanged.
2. **The propagator question.** In the stationary bulk, Q_t is (asymptotically) Toeplitz, so consecutive rows of the score operator are shifts of one another — the formal skeleton of the propagator hypothesis $S_{k+1} \approx P_t S_k$. The bulk analysis of §11 (the q-geometric structure of J^{-1}, hence of Q_t by the same transfer-matrix argument) is the right starting point for making P_t explicit and quantifying its boundary corrections.
3. **Local score architectures.** Theorem 19 gives a principled radius: $r \approx \xi \log(1/\varepsilon)$ with $\xi = 1/\log(1/q(\alpha, t))$ — a noise-level-dependent receptive field. It would be instructive to test whether trained video-score networks develop receptive fields tracking $\xi(t)$.
4. **Beyond scalar frames.** For $a_k \in \mathbb{R}^d$ with linear dynamics, every formula survives with $\alpha \to A$ (a matrix) and Kalman algebra; the bulk analysis becomes a matrix quadratic — worth writing once a concrete $d > 1$ toy (e.g. the rotating frame) is fixed.

16 Reproducibility

Everything in this note is reproducible from the repository `GioviManto/gaussian-bp-diffusion`:

- `code/ar1_utils.py` — model: Σ_0, Q_0, Σ_t, Q_t (direct and spectral), exact score (matrix and Tweedie), posterior.
- `code/bp_score.py` — Convention-A BP: forward / backward sweeps, combination, BP score.
- `code/chain_formulas.py` — every closed form of §11–12: J_d, β, λ^*, V_exact, q, bulk covariance, V_AMP, existence, t_c, α_c, weak-coupling error; and the $K = 2$ forms of §6.
- `code/amp.py` — the three marginal schemes (mean field, AMP/TAP, exact) on arbitrary (J, h); honest breakdown reporting.
- `code/local_bp.py` — radius-r local estimator and the exact RMS truncation error.
- `code/experiments.py` — all figures.
- `code/numerical_audit.py` — the gate: **72 checks**, all passing.
- `01_model_and_score.ipynb`, `02_belief_propagation.ipynb`, `03_amp_and_experiments.ipynb` — executed companions mirroring §2–6, §7–10, and §11–14 respectively.

To reproduce from scratch:

```
python3 -m venv .venv && source .venv/bin/activate
pip install numpy matplotlib jupyter nbclient nbformat
python code/numerical_audit.py    # -> PASSED 72 / 72
python code/experiments.py        # -> figures/*.png
python build_companions.py        # -> the three executed notebooks
tectonic main.tex                 # -> main.pdf
```

A The numerical audit, section by section

The audit philosophy: *a statement enters the text only if an independent check passes.* Independent means: the closed form is compared against a different computational route (brute-force inversion, spectral identity, bisection of an iteration, a fitted slope), never against itself. The 72 checks group as follows; worst errors are over all configurations in each group.

§	contents	checks	worst error	tolerance
1	$Q_0 = \Sigma_0^{-1}$ exactly tridiagonal (12 configs)	24	1.8×10^{-14}	10^{-12}/exact
2	$\Sigma_t Q_t = I$; spectral form; U diagonalises Σ_0	15	2.2×10^{-14}	10^{-10}
3	band-fill coefficient, $d = 1..5$ (no factorial)	5	9.0×10^{-3}	5×10^{-2}
4	large-t return $\|Q_t - I\|_F \sim e^{-2t}$	3	2.1×10^{-2}	struct.
5	BP vs matrix: mean / variance / score (80 configs)	3	1.2×10^{-14}	10^{-10}
6	Tweedie route = matrix route	1	2.7×10^{-15}	10^{-12}
7	AMP mean = exact; AMP score = BP score (36 configs)	2	4.3×10^{-12}	10^{-9}
8	AMP variance: dense-graph accuracy; chain breakdown	4	1.7×10^{-3}	5×10^{-2}
9	local BP: $r{=}K{-}1$ exact; error monotone in r	2	3.6×10^{-15}	10^{-9}
10	$K = 2$ closed forms: Σ_t, Q_t, score	3	1.8×10^{-15}	10^{-12}
11	bulk closed forms: V_{exact}, recombination and existence, V_{AMP}, boundary scan, $q^{\,d}$ covariance, weak-coupling ratio, locality slope, t_c, α_c	10	3.4×10^{-3}	varies

Run `python code/numerical_audit.py` to regenerate the full 72-line report; each line names the claim, the measured error, and the tolerance.

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